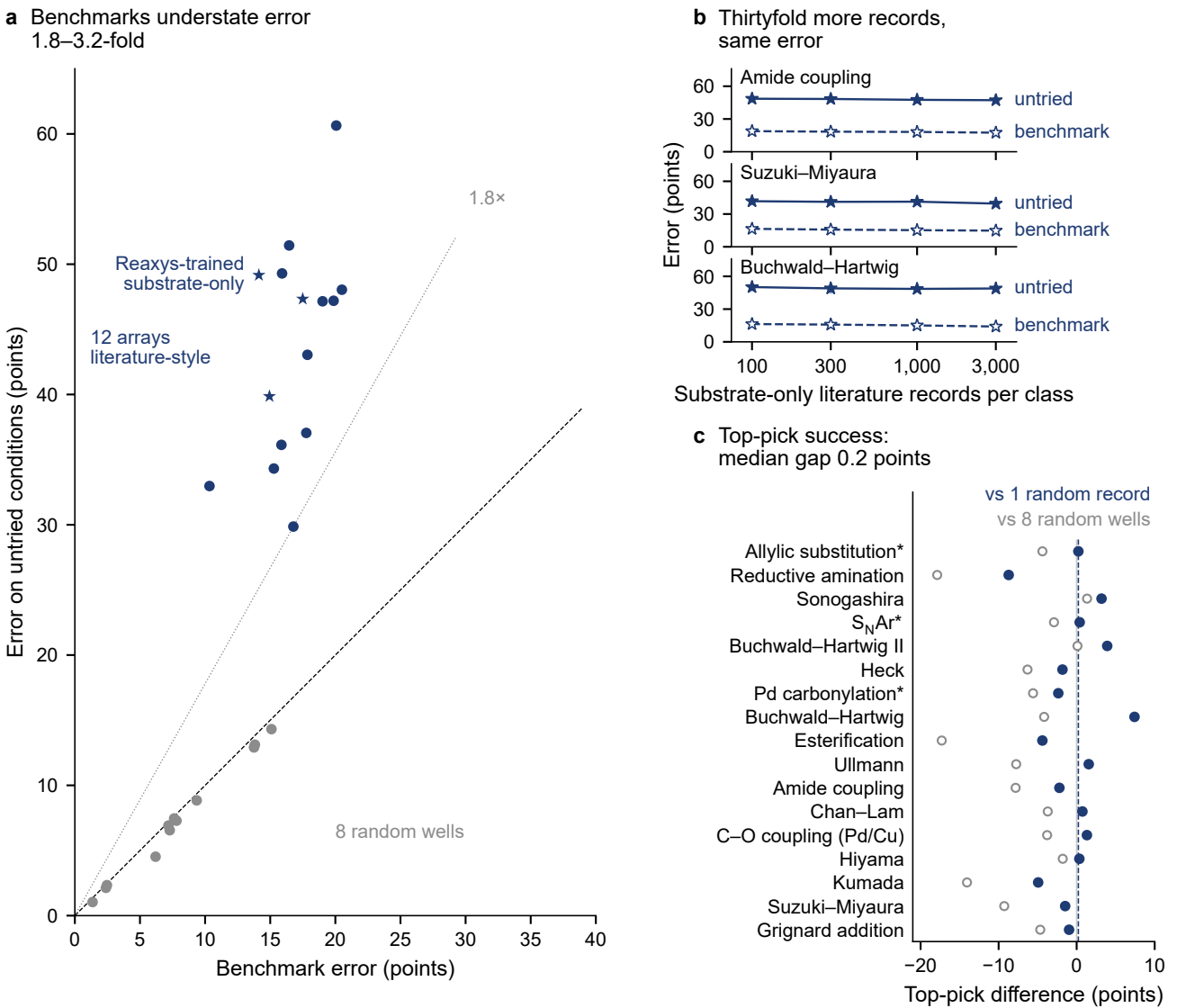
a Benchmarks understate error
1.8–3.2-fold
Error on untried conditions (points)
Benchmark error (points)
1.8×
Reaxys-trained
substrate-only
12 arrays
literature-style
8 random wells
b Thirtyfold more records,
same error
Amide coupling
Suzuki–Miyaura
Buchwald–Hartwig
untried
benchmark
Error (points)
Substrate-only literature records per class
c Top-pick success:
median gap 0.2 points
vs 1 random record
vs 8 random wells
Allylic substitution*
Reductive amination
Sonogashira
SNAr*
Buchwald–Hartwig II
Heck
Pd carbonylation*
Buchwald–Hartwig
Esterification
Ullmann
Amide coupling
Chan–Lam
C–O coupling (Pd/Cu)
Hiyama
Kumada
Suzuki–Miyaura
Grignard addition
Top-pick difference (points)